Hierarchical designing → decomposing function into small Blocks

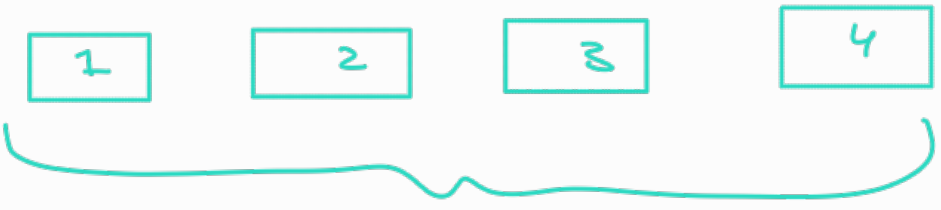
→ quicker to use

To deal with complexity of a circuit it is broken into blocks. The blocks are interconnected to form a circuit.

E.g

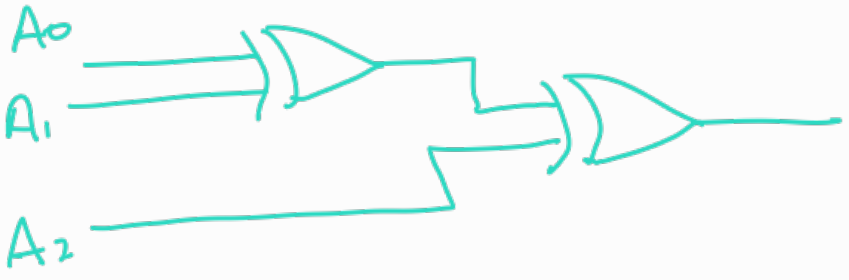
if we are required 20 AND gates.

we suppose a block consisting of 5 AND gates in it solve the problem.



20 AND gates

e.g Design a 9 input odd function for checking even parity



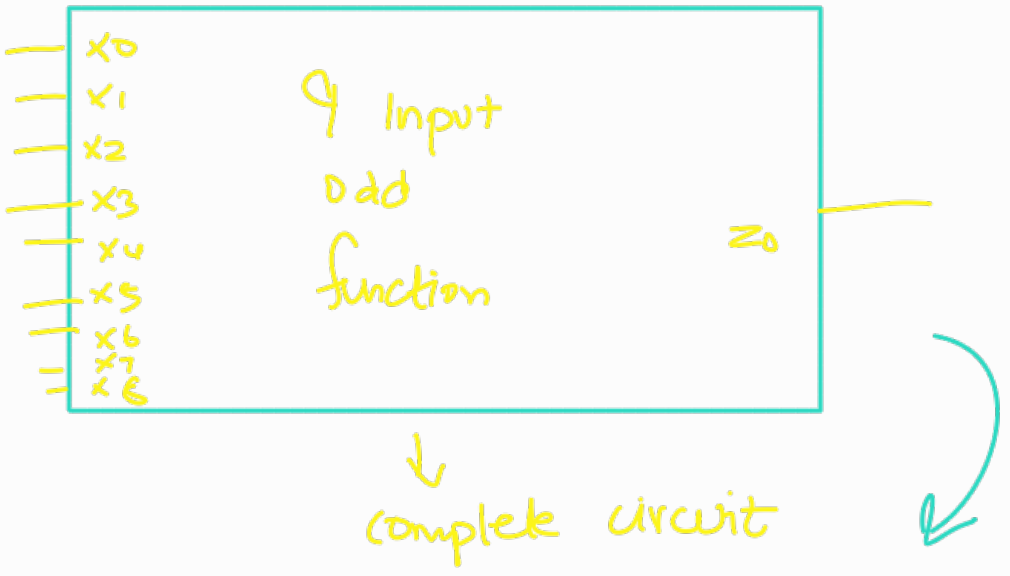
3-input odd function.

we make a block for each 3 input odd function.

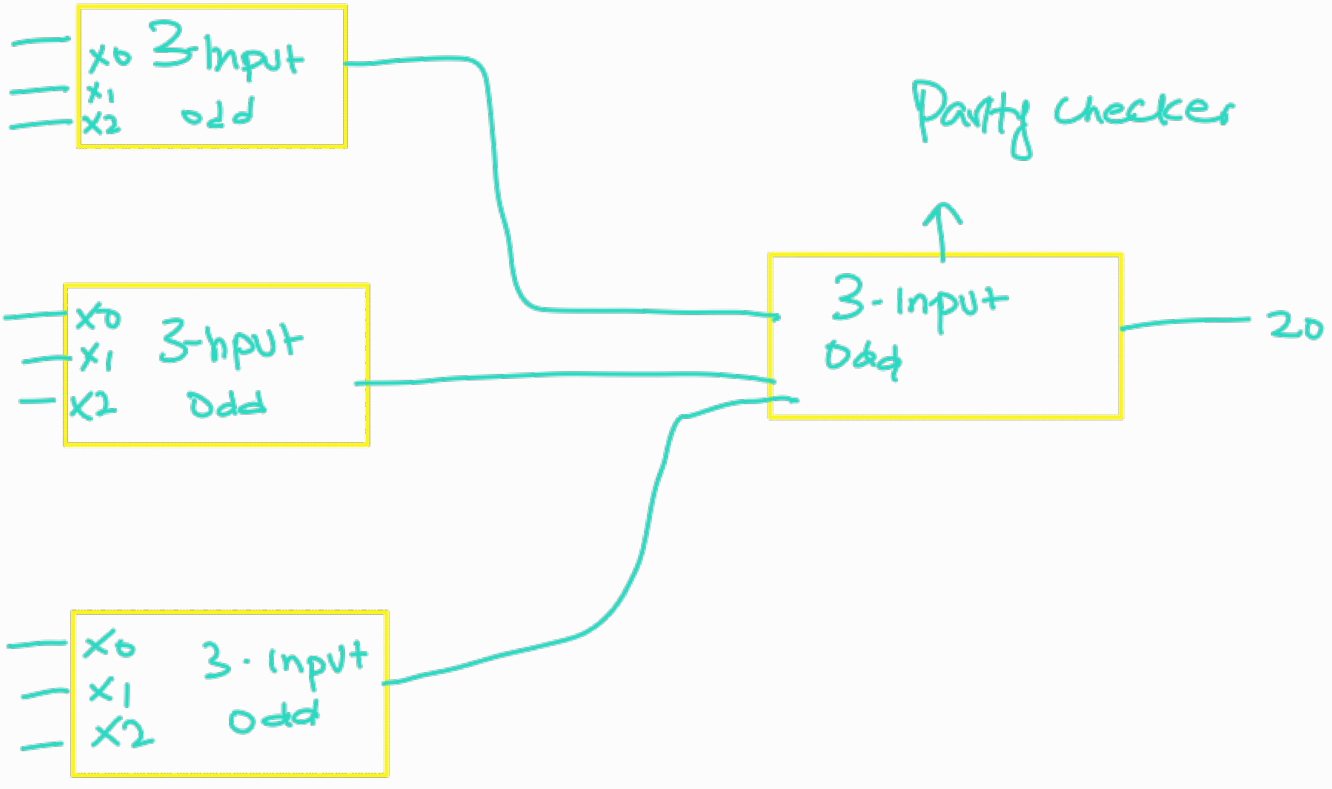


Block for 3-input odd function

Now three such blocks are needed and one output block to detect parity.



↓ complete circuit



Equality comparator:. To determine whether every bit of both numbers are equal or not.

It only gives 1 output if A=B

Magnitude comparator :- To determine A and B and their relative magnitudes.

The outcome is based on three outputs.

i) $A > B$

ii) $A = B$

iii) $A < B$



Draw t.T for A and B each having 1 bit compare A with B

no of possible combinations for two n bit no's

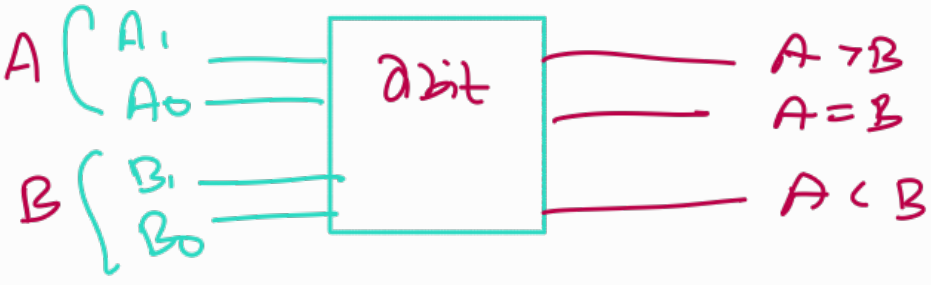
$$2^{2n} = 2^{2(1)} = 4$$

A	B	A<B	A>B	A=B
0	0	0	0	1
0	1	1	0	0
1	0	0	1	0
1	1	0	0	1

3 - kmaps will be made.

2 bits comparator circuit.

Comparision can be made by two numbers only. But now each of the two numbers A and B have 2 bits. Hence $2^{2(2)} = 16$ possible outcomes





A_1	A_0	B_1	B_0	A>B	A<B	A=B
0	0	0	0	0	0	1
0	0 → 0	0	1 → 1	0	1	0
0	0 0	1	0 2	0	1	0
0	0	1	1	0	1	0

0 3

use decimal equavient to compare

Q Design a 4-bit equality comparator using hierarachial method.

Meaning $\underbrace{A_3 A_2 A_1 A_0}_{A}$ and $\underbrace{B_3 B_2 B_1 B_0}_{B}$

Logic if A and B are equal $E=1$ (output)

if A and B are unequal $E=0$

Total inputs $2^{2(4)} = 256$ possible.

X very complex.

we can decompose this problem into four 1 bit comparision circuits (MX) and a additional circuit (ME) that combines the four comparision circuits outputs to obtain 'E'.